

The Wisdom of Crowds vs. Statistical Models

Measuring Market Efficiency in NBA Outcome Predictions

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Background:

Prediction markets such as Kalshi allow participants to trade binary contracts on the outcome of thousands of events happening around the world, including professional basketball (NBA) games. The current price, to buy or sell, one of these contracts ("Will __ team win?") essentially reflects the wisdom of the crowd at any given moment as all available information is being factored into that price. These market-implied probabilities are great for data analysis and can be used to compare against statistical models.

Our project seeks to evaluate the closing-line price from the Kalshi prediction market website against a simple ELO rating system for every regular-season game during the 2025-2026 NBA season. What we'd like to learn is if the market has done its job of pricing in all the statistical cues from current season performances, or if there are any lingering systematic biases that a model driven by the numbers can take advantage of.

Motivation:

Prediction markets have recently been gaining a reputation as a very efficient aggregator of public and private information. Our theory is that if these markets are so efficient then the crowd should have already incorporated all the available statistical information into the contract prices. So, the central question we seek to address in our project is; do these markets actually fully incorporate statistical signals from performance data or are there systematic biases that exist which can be exploited by data-driven models. Overall, the goal is to quantify the calibration error of the market and determine whether a statistical model can identify consistent edges relative to market prices.

Questions:

- 1) Which forecaster is more accurate - the Kalshi market or an ELO model (based on Brier score)?
- 2) Is the market well calibrated - does an event priced at 70% actually occur ~70% of the time?
- 3) When the model and the market disagree, who turns out to be right?

Data Sources - NBA Games and Kalshi Market Odds

Title	Description	Size	Location	Format	Access
NBA Regular Season Schedule & Results (2025 - 2026)	The Sports Reference website keeps a detailed log of every NBA game that's occurred throughout the season. This was our primary source and included: game-by-game schedule, final scores, dates, start time (ET), visitor/home teams, visitor/home points, arenas, and attendance. We used this dataset throughout the rest of the notebook including for: fetching Kalshi price data, team ELO ratings, and final accuracy metrics.	1,232 rows	Basketball-Reference (Sports Reference) website	CSV	Download (nba_reg_2526.csv)
Kalshi NBA Game Markets (2025 - 2026)	Our secondary dataset consists of trade data from the Kalshi prediction market API. We used specific endpoints and data from our primary source to extract the closing-line odds as the 'yes_price_dollars' of the earliest trade occurring at or immediately after the scheduled tip-off time. This approach was the best way to capture the market consensus immediately before the game began. The dataset was accessed programmatically via the Kalshi Developer API and returned in JSON format before being parsed into a pandas DataFrame.	1,231 games queried; thousands of underlying trades (JSON responses)	Kalshi Developer API (trades endpoints)	JSON	API retrieval via Python requests

Data Manipulation: NBA Schedule Cleaning

We created the custom 'nba_season_converter' function to transform our raw game csv file into an analysis ready pandas DataFrame.

DateTime Consolidation:

Our first step in the function, after importing the data / creating the DataFrame, was to consolidate the date and time columns into a single 'DateTime_ET' column. From the raw csv, for example, the date ("Tue Oct 21 2025") and start time ("7:30p") are stored as separate strings in separate columns. So, what we did was combined these two columns, replaced the "p"/"a" suffixes with "PM"/"AM", and then parsed each row with `.to_datetime()` using an explicit format (%a %b %d %Y %I:%M %p).

GameID Engineering:

The next step was to create our 'GameID' column which was engineered to precisely match Kalshi's market ticker format. Having accurate market tickers for each game is crucial as later we will use these to fetch trade data (being even slightly off will result in code bugs or missing data). And we did this by combining the parsed datetime (year, month, day) with mapped home team codes. To get the home team codes, we used the 'team_to_code' dictionary, which maps each city and team name from the raw csv to Kalshi ticker abbreviations ("Oklahoma City Thunder" -> "OKC").

Winner & TimeStamp Columns:

Lastly, our raw csv contained no explicit "Winner" column so we had to create one by simply comparing the visitor and home points. And additionally, we created a 'TimeStamp' column which is just a Unix TimeStamp for game start times (which will also be used later to fetch Kalshi closing-line trades).

Cleaning produces three DataFrames

full_nba_season_df

Cleaned full-season results — the base for the final merge

historical_games

Oct 2025 – Mar 2026 games → Kalshi historical trades endpoint

april_games

Apr 2026 games → Kalshi recent-trades endpoint (lag cutoff)

Data Manipulation: Kalshi Odds & Final Merge

Kalshi API Endpoints:

We had to create two key functions for fetching Kalshi trade data. Kalshi's "historical/trades" endpoint has a lag period of exactly two months, so we were able to use it in the 'kalshi_historical_cl_prices' function to get trade data from March 2026 to basically any prior date. However, for anything after March, which includes the last few regular season games in April we had to use 'kalshi_april_cl_prices' function with the "markets/trades" endpoint. In the Kalshi API, this endpoint is where trades are kept essentially in a limbo state (they're closed for trading but have not yet been officially moved to historical data). Also, it's important to note that if our notebook is ran anytime after June 12th 2026, only the 'kalshi_historical_cl_prices' function is needed as the last games of the regular season were April 12th 2026.

Closing Line Trade Prices:

For both functions mentioned above (the rest of the code is exactly the same), after the Kalshi endpoint was hit we then parsed the JSON response for all trades that happened from tip-off to 550 seconds after. We do this in order to guarantee there is at least one trade for every game as quite often there'd be no trades that happened at the exact moment of our TimeStamp (game start). We then filter to the earliest returned trade by "created_time" and use the "yes_price_dollars" as the market's closing-line home-win probability (games with no trades in the window became NaN). We also had to use a sleep delay between requests in order to respect Kalshi's API rate limits (which is why the process takes so long ~15 minutes).

Merge & Final Cleaning:

Lastly, we then concatenate the April prices DataFrame with the historical prices DataFrame to get a one fully combined Kalshi price DataFrame (this is skipped if code is ran after June 12th). Then we use the 'final_combined_df' function to merge this with our original cleaned full-season NBA DataFrame, drop any NAN rows (2 in our case), calculate the visitor odds (by simply just subtracting 1 from the home price odds), drop/rename/reorder columns, and then export the final result as 'final_combined.csv' which is ready for the analysis stage.

full_nba_season_df



+ Kalshi closing prices
(historical U April endpoints)



dropna · rename · reorder



final_combined.csv
1,229 games × 9 columns

Statistical Baseline: The ELO Rating Model

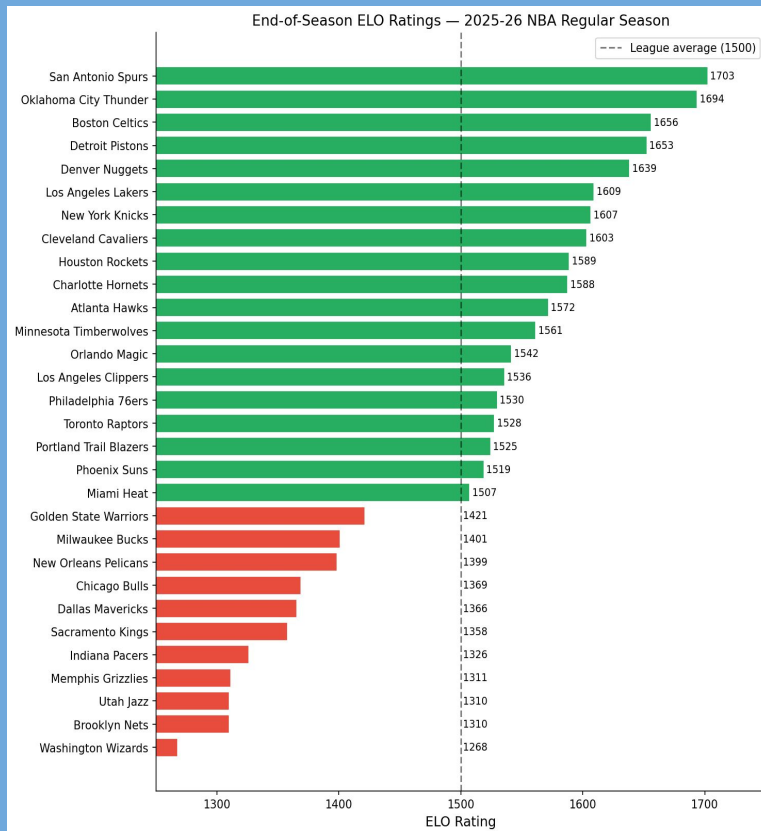
Model Setup:

The ELO model is implemented by having all 30 NBA teams at the start of the season begin with a rating of 1500 (league average). A fixed +75 bonus is always applied to the home team when computing pre game win probabilities (equivalent to about a 60.7% home-win probability for two perfectly matched teams). Then after every game, the winner gains while the loser loses points, in proportion with how surprising the result was and with K-factor = 20 controlling the update size. We used this system to calculate the ELO ratings for each team throughout all 1,229 games and in chronological order. As well, for every game in the 2025-2026 season we stored the pre game ratings for both teams and the model's home win probability.

Win Probability Formula

$$P(\text{home wins}) = \frac{1}{1 + 10^{(ELO_{\text{vis}} - (E_{\text{ome}} + 75))/400}}$$

When doing a final comparison of the ratings from the ELO model to season outcomes, it suggests that the model actually produced pretty reasonable estimates of team strength. For example, the Spurs and Thunder at the end of the season were considered to be real championship contenders and both teams had deep playoff runs. This pattern is a strong indicator that our ELO model was successful in differentiating between strong and weak teams throughout the season.



Analysis: Brier Score — Who Is Closer to the Truth?

Method:

The Brier score measures the accuracy of probabilistic predictions. It evaluates how close a predicted probability is to the actual outcome by calculating the mean squared difference between the two. A home win is coded as 1 and a loss as 0.

Lower values indicate better predictive performance. As a baseline, a naive model that always predicts a 50% probability for each game achieves a brier score of 0.25, which serves as the benchmark that any useful model should outperform.

$$BS = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - o_i)^2$$

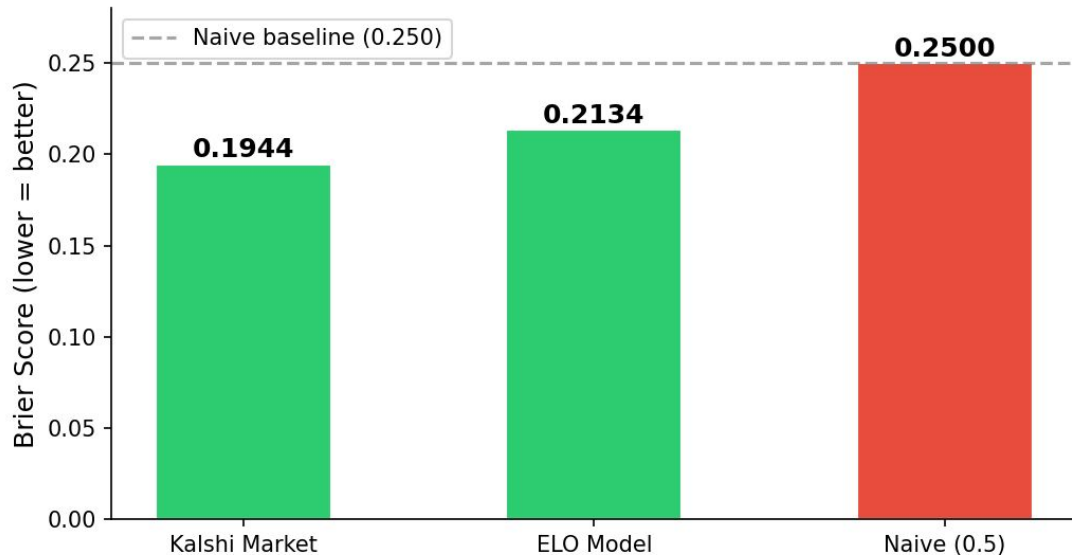
Result:

Both the Kalshi market data and ELO model beat the naive baseline, but the Kalshi market also outperformed the ELO model by roughly 0.019 Brier points. This is a significant gap given the large sample size of 1,229 games.

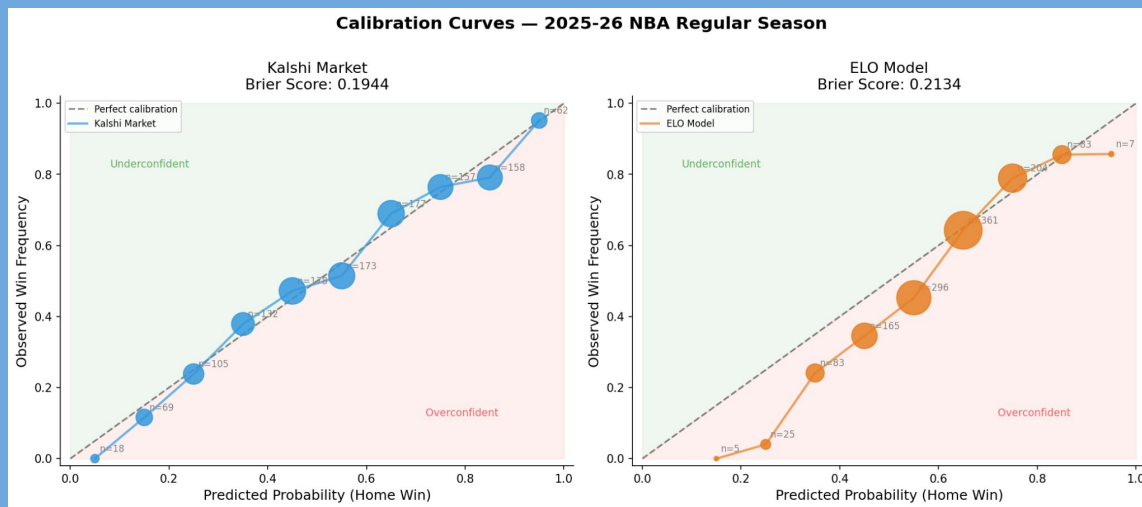
Interpretation:

The crowd's prices contain real information beyond simply just win loss history - such as injuries, rest, lineups, and matchups that a pure ELO model does not. This is a strong piece of evidence consistent with an efficiently priced Kalshi market.

Brier Score: Kalshi Market vs. ELO Model



Analysis: Calibration Curves (Reliability Diagrams)



Method:

A calibration curve, tests whether predicted probabilities are honest by first grouping games into their probability bins and then comparing the average predicted probability against the actual observed win frequency for each bin. We see perfect calibration happen when there are points lying exactly on the diagonal line. When points are below the diagonal it indicates overconfidence and when points are above the diagonal it indicates underconfidence.

Result:

We found that the Kalshi market stays pretty close to the diagonal across the full probability range, meaning that games priced at 70% won roughly 70% of the time. The ELO model, however, did not do as well and is significantly below the diagonal line in the 0.4–0.7 range. Meaning that it systematically is overconfident in home teams (it predicts home wins more often than they occur). Furthermore, when weighted by bin counts we found that the market's mean calibration error is about 3 percentage points versus roughly 6 percentage points for ELO.

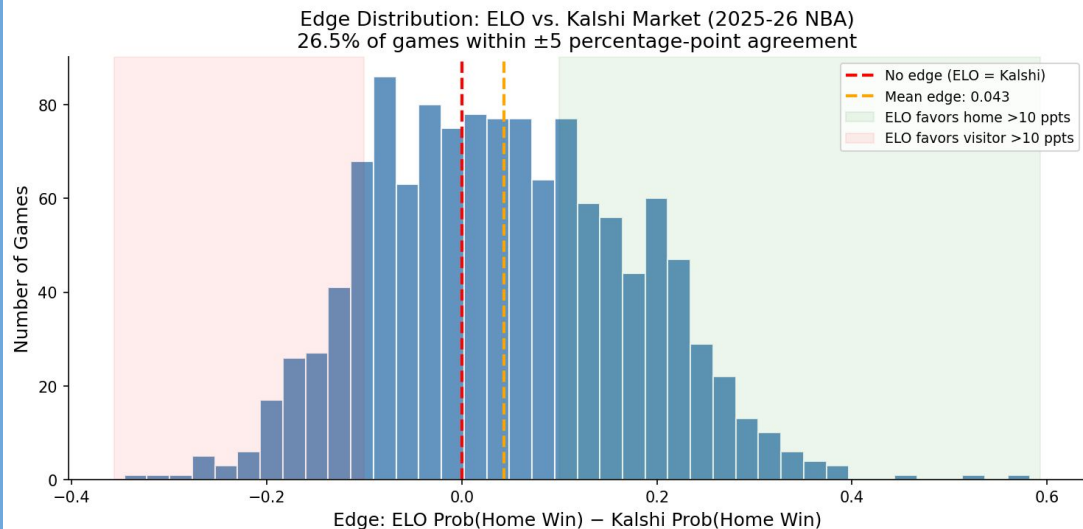
Analysis: Edge Distribution (Model vs. Market)

Method:

We define "Edge" here as simply the difference between "ELO Home Win Probability" and "Kalshi Home Win Probability" for each game. So, when our ELO model had a higher win probability for the home team in comparison to the Kalshi market home win probability, we get a positive edge (and a negative edge for when it was lower).

Formula:

Edge = ELO P(Home Win) - Kalshi P(Home Win)



Results:

Across the 1,229 games we got that the mean edge was +0.0428, meaning that our ELO model gave home teams about 4.3 percentage points higher of a win probability than Kalshi did on average. As well, we found that the median edge was +0.0385 and the standard deviation was 0.1311 with an overall distribution ranged from -0.346 to +0.583 (as displayed in the histogram). Furthermore, we found that nearly 26.5% of games fall within ± 5 percentage points of agreement, and nearly half diverge by more than 10. Overall, the distribution is right-shifted, which is consistent with the calibration finding that ELO's fixed +75 home-court bonus overrated home teams in a season where home teams won just 55.5% of games.

Limitations & Ethical Considerations

Limitations

There are several limitations that should be kept in mind when interpreting the results from this project.

Single Season:

Firstly, the analysis covers only a single season (2025 - 2026) so there's definitely a possibility that the conclusions are unrepresentative for prior years. To fully test the robustness across different league conditions and team compositions, multi-year data would be a necessary.

Simple Baseline Model:

Furthermore, our analysis model (ELO Model) that we used was deliberately kept on the simpler side. Elo is calculated using very limited number of variables (wins, losses, and a fixed home bonus) while also ignoring several important ones (injuries, rest, travel, and roster changes, trades, etc).

Closing-Line Price Probabilities:

In addition, our closing-line prices were taken from the first trade at the start or right after tip-off; in low-volume games this has the potential to create some noise in the data.

Kalshi Price Fee Structure:

Lastly, in real trading Kalshi charges additional fees for users depending on the trade type (limit orders have no fees but market orders have a varying fee structure). None of these fees were factored in to our price probabilities as that information is unavailable.

Ethical Considerations

Responsible Gambling Framing:

Our project is meant to be used for educational purposes only and study the market efficiency for NBA games. Our findings should not be taken as betting advice of any kind or be used to encourage irresponsible gambling. Our focus throughout has been about understanding information aggregation and statistical calibration rather than on strategies for beating the bookmakers.

API Responsibility:

Every API call that we made through this project has respected Kalshi's published rate limits. We have taken deliberate steps in the writing of several of our functions to implement sleep delays between requests. As well, data pulls were batched and cached to CSV to avoid repeated load.

Public Data Only:

All the data sources we've used in this project are publicly available. There's been no personal or proprietary data used of any kind. All trade records pulled from the Kalshi API are anonymous.

Summary & Opportunities for Further Research

Q1: The Crowd Wins

Both the Kalshi market and our basic ELO model beat the Naive baseline of 0.250, however it was the Kalshi market that won overall with the lowest Brier score of 0.1944 compared to the ELO model at 0.2134. These results support the idea that the crowd aggregates all available information effectively (injuries, rest, lineups, etc) and will perform better especially when in comparison with the limited ELO model.

Q2: The Market Keeps its Promises

The Kalshi prediction market for NBA games demonstrates high but not perfect accuracy. Its probabilities are overall better calibrated, as we mentioned before and in a large part due to its ability to incorporate all available information. Market probabilities matched observed frequencies very closely across all deciles (~3% mean error). Events priced at 70% happened about 70% of the time. In contrast, our ELO model seemed to be systematically overconfident in home teams (~6% mean error) which is likely due to several of factors and in its simplicity.

Q3: Edges were Illusions

We found that our ELO Model and the Kalshi market disagreed quite frequently (about 47.6% of games diverged by >10 pts) creating opportunities for there to be an “Edge”. However, we found absolutely no evidence of any type of systematic bias that this simple ELO model could exploit. Essentially, it turns out that going with the efficient market is the right choice when the two are not aligned.

Further Research

- 1) Expand the data collection process to include multiple seasons, playoff games, and even international leagues. With abundantly more data it's logical to assume that more accurate conclusions will be made.
- 2) Use the complete Kalshi order-book and include multiple trades per game (instead of just the closest to the start time). This will lead to more accurate closing-line prices and potentially better results.
- 3) Improve the statistical model; using essentially a publicly available ELO ratings model is highly unlikely to beat the Kalshi market prices. Creating a more sophisticated model that incorporates richer features or access to real-time information has the potential to actually find meaningful discrepancies.

Statement of Work & References

Statement of Work

Bradley Hofmeister

Led the writing of code for the NBA schedule cleaning function. As well as the Kalshi API integration; the functions for the endpoint trade retrieval, JSON-to-DataFrame parsing, closing-line price extraction, and the merged pipeline producing the 'final_combined.csv'. Primary author of Part 1 of our 'Milestone1_Project.ipynb' notebook.

Dominick Devarti

Led the writing of code for the ELO model implementation, the Brier analysis, the calibration analysis, the Edge analysis and all the visualizations. Primary author of Part 2 of our 'Milestone1_Project.ipynb' notebook.

Shared

Both members contributed in the overall direction of the project, the computational narrative, the writing/editing of the 'Milestone1_Project.ipynb' notebook, and the writing and editing of this final report.

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